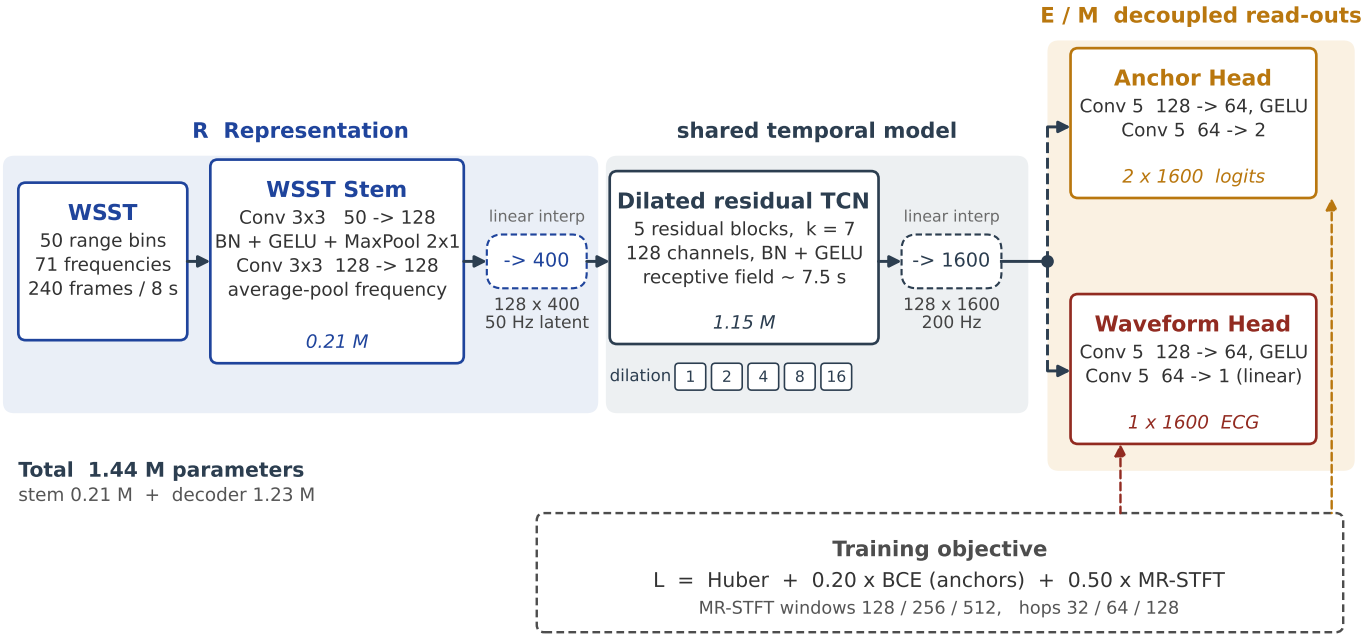
E / M decoupled read-outs
R Representation
shared temporal model
WSST
50 range bins
71 frequencies
240 frames / 8 s
WSST Stem
Conv 3x3 50 -> 128
BN + GELU + MaxPool 2x1
Conv 3x3 128 -> 128
average-pool frequency
0.21 M
linear interp
-> 400
128 x 400
50 Hz latent
Dilated residual TCN
5 residual blocks, k = 7
128 channels, BN + GELU
receptive field ~ 7.5 s
1.15 M
dilation 1 2 4 8 16
linear interp
-> 1600
128 x 1600
200 Hz
Anchor Head
Conv 5 128 -> 64, GELU
Conv 5 64 -> 2
2 x 1600 logits
Waveform Head
Conv 5 128 -> 64, GELU
Conv 5 64 -> 1 (linear)
1 x 1600 ECG
Total 1.44 M parameters
stem 0.21 M + decoder 1.23 M
Training objective
L = Huber + 0.20 x BCE (anchors) + 0.50 x MR-STFT
MR-STFT windows 128 / 256 / 512, hops 32 / 64 / 128